# VISVESVARAYA TECHNOLOGICAL UNIVERSITY
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**A Major Project Phase – II (BIC786)**

**Report on**

# "BovineGuard: IoT-Driven Cattle Wellness Tracker"

*Submitted in partial fulfilment of the requirements for award of degree*

of

**BACHELOR OF ENGINEERING**

In

**CSE (IoT & CSBT)**

**By**


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# CERTIFICATE

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It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

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# DECLARATION

We hereby declare that the project report entitled **"BovineGuard: IoT-Driven Cattle Wellness Tracker"**, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering (IoT & CSBT) of Visvesvaraya Technological University, Belagavi, is a record of our original work carried out by us under the supervision of **Mr. Jay Prakash Malviya**, Assistant Professor, Department of **CSE (IoT & CSBT)**, EPCET.

We further declare that this project report has not been previously formed as the basis for the award of degree. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

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# ABSTRACT

**BovineGuard** is an Internet of Things (IoT)-driven system engineered for the persistent, real-time health surveillance of cattle. Developed to solve the critical issue of belated illness identification in livestock management, the core system employs an **ESP32 Microcontroller** connected to three key sensors: the **DS18B20** for accurate body temperature, the **MAX30100** for heart rate monitoring, and the **MPU6050** for tracking physical activity and motion. Data collected is processed locally against established health thresholds. When a critical deviation is detected (e.g., fever or prolonged inactivity), the system immediately engages a **GSM module** (SIM800L) to transmit urgent **SMS notifications** directly to the farm manager or veterinarian. This design strategically avoids dependence on internet connectivity, making the solution viable and robust in remote, rural settings. The subsequent development phase incorporates a **mesh network** to centralize monitoring from numerous animals onto a single GSM node, bolstering affordability and large-scale deployment. BovineGuard enables swift intervention, enhancing animal welfare and supporting precision livestock farming.

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# Abbreviations/ Notations/ Nomenclature (if any)

| No. | Abbreviation/Notation | Full Form/Description |
|---|---|---|
| 1 | GPRS | General Packet Radio Service |
| 2 | GSM | Global System for Mobile Communications |
| 3 | WSN | Wireless Sensor Network |
| 4 | MCU | Microcontroller Unit (refers to ESP32 S3) |
| 5 | BPM | Beats Per Minute (Unit for Heart Rate) |
| 6 | I2C | Inter-Integrated Circuit (Communication Protocol) |
| 7 | UART | Universal Asynchronous Receiver-Transmitter |
| 8 | GPIO | General Purpose Input/Output |
| 9 | SMS | Short Message Service |
| 10 | PCB | Printed Circuit Board |

# INTRODUCTION

# Chapter 1

# INTRODUCTION

The confluence of agricultural technology and embedded systems offers unparalleled opportunities to redefine traditional farming practices, particularly in the critical domain of livestock management. The global economic viability and ethical standards of cattle farming are intrinsically linked to the sustained health and prompt welfare intervention for the herd. To decisively address the systemic limitations inherent in conventional, labor-intensive inspection regimes, this project introduces BovineGuard, an innovative, fully autonomous Internet of Things (IoT)-based solution.

BovineGuard is meticulously engineered to provide continuous, real-time surveillance of essential physiological and behavioral indicators. By deploying advanced micro-sensors and implementing localized processing capabilities on the device, the system guarantees that any measurable deviation from normal health parameters is detected immediately. This, in turn, triggers a rapid and reliable notification mechanism that is purposefully independent of fixed communication infrastructure, making the solution universally applicable. The development of BovineGuard is fundamentally centered on enhancing animal welfare, minimizing mortality rates, and promoting the adoption of precision livestock farming techniques in all environments, especially those geographically isolated or lacking robust cellular coverage.

## 1.1 Background

Modern industrial livestock production, particularly within the cattle sector, is continually challenged by the need to ensure animal welfare while maximizing productivity. Traditional health surveillance methods, which rely primarily on periodic manual observation by farm staff or veterinary checks, are fundamentally prone to significant latency in detecting the early onset of illness. This delay often translates directly into extended periods of suffering for the animal, increased rates of preventable mortality, and substantial financial repercussions for the operator due to reduced performance, necessary isolation, or the cost of prolonged intensive treatments. The technological shift enabled by

the Internet of Things (IoT) presents a paradigm-altering solution. By integrating low-power microcontrollers, such as the **ESP32**, with specialized bio-sensors (e.g., DS18B20, MAX30100, MPU6050) directly into a wearable monitoring unit, continuous streams of vital health data can be acquired. This technological intervention effectively moves the process from reactive illness treatment to a proactive health maintenance strategy, ensuring notification at the first objective sign of physiological distress. The BovineGuard project is anchored in the application of this embedded IoT methodology to create a robust, cost-accessible, and highly scalable monitoring solution tailored for the specific demands of remote agricultural terrains.

## 1.2 Problem Statement

The successful and efficient management of a livestock operation is inextricably linked to the ability to swiftly identify and isolate sick animals. The foremost obstacle to optimizing cattle health management remains the critical time lag between the pathological manifestation of a disease and its definitive diagnosis. Early indicators of systemic issues—such as an elevated core body temperature (fever), a significant decline in activity levels, or an irregular heart rate—are often too subtle to be consistently identified during routine, intermittent manual checks. Furthermore, for large-scale farm operations spanning vast geographical areas and monitoring potentially hundreds of animals, continuous, individualized physical observation is both economically prohibitive and logistically unsustainable. A parallel problem lies with many existing commercial electronic monitoring systems, which mandate reliable high-speed internet connectivity (e.g., Wi-Fi, 4G/5G) for the transmission of data to cloud-based servers. This reliance renders them fundamentally impractical for the majority of rural and remote farming locations that inherently lack adequate digital infrastructure. Consequently, the absence of a reliable, real-time, and infrastructure-independent alerting mechanism directly contributes to preventable cattle mortality and the sub-optimal, widespread use of prophylactic medication. The central problem addressed by BovineGuard is thus the necessity for an **affordable, autonomous, and communication-agnostic embedded system** that delivers immediate, location-aware alerts to farm personnel upon the detection of critical health deviations in any monitored animal.

## 1.3 Existing System

Current standard practices for livestock health management are dominated by manual and subjective techniques, primarily relying on routine visual inspection by farm personnel. This approach requires observers to identify overt, visible symptoms of distress, such as pronounced lethargy, severe changes in feeding habits, or obvious physical abnormalities. This methodology is fundamentally constrained by being highly labor-intensive and reliant on the skill and constant presence of the observer, making it effective only after an illness has progressed to a visually unmistakable stage. Given the immense scale of modern commercial ranching, consistent, individualized animal surveillance becomes logistically impractical. Consequently, diseases often escalate in severity or spread undetected within the herd, forcing the adoption of reactive, costly treatments that generally yield a lower success rate compared to strategies based on preventative or early intervention. This legacy system critically lacks the capability for non-invasive, objective, and early-stage physiological detection.

Although various electronic monitoring systems have emerged to mitigate the inherent limitations of manual checks, a significant majority of these solutions suffer from constraints that critically impede their widespread, practical adoption in remote agricultural settings. The foremost technical barrier is the absolute requirement for consistent, high-bandwidth internet connectivity for transmitting large volumes of sensor data to a centralized processing server or cloud platform. For farms situated in rural, mountainous, or geographically isolated terrain, establishing and maintaining the requisite network quality is frequently cost-prohibitive or technically impossible. Furthermore, many current commercial offerings utilize closed, proprietary hardware architectures and necessitate costly centralized hub equipment. This structure drastically inflates the initial capital expenditure per animal, severely undermining the project’s necessary goals of affordability and scalable deployment, particularly for small and mid-sized farming enterprises. BovineGuard is specifically designed to overcome these shortcomings by centering its core system on an inexpensive, non-proprietary hardware stack and prioritizing the **GSM-based communication protocol**, offering a genuinely robust and accessibility-focused alternative.

## 1.4 Proposed System

The BovineGuard system is conceptualized as a dedicated, fully self-contained IoT node engineered for continuous physiological and 5ehaviour5l surveillance of cattle. The foundational unit utilizes the ESP32 S3 microcontroller, selected for its optimal balance of powerful processing capabilities, minimal power draw, and built-in wireless functionality essential for network growth. The ESP32 S3 serves as the primary processing hub, executing firmware that receives, interprets, and runs diagnostic analysis on data acquired from a suite of biomedical sensors mounted on the animal.

Vital signs data is meticulously collected through a tri-sensor array. Core body temperature, a key indicator of systemic infection or fever, is precisely measured using the DS18B20 digital temperature sensor. The physiological activity of the animal, specifically cardiac rhythm and pulse rate, is non-invasively monitored by the MAX30100 pulse oximetry sensor, allowing for the detection of stress or cardiac anomalies. Finally, physical activity, encompassing movement patterns, rest durations, and identifying unusual, static 5ehaviour (a strong indicator of distress), is tracked using the MPU6050 six-axis accelerometer and gyroscope. This multi-modal sensing approach provides the most comprehensive and redundant physiological profile available.

Data analysis is performed entirely on-device using sophisticated, firmware-embedded threshold-based logic. Once the processed data from any sensor exceeds or drops below established healthy parameters, the system autonomously classifies the event as an abnormal condition. The architectural design is structured to prioritize alert levels: Non-Critical (status updates) and Critical (emergency). Critical alarms trigger the immediate activation of the integrated GSM module (SIM800L). This module ensures rapid, infrastructure-independent communication by generating and dispatching an urgent SMS notification containing the animal’s unique ID and the specific nature of the health emergency directly to pre-programmed contacts. This fundamental design choice ensures the system’s operational continuity and maximum effectiveness, regardless of the surrounding digital communication infrastructure.

## 1.5 Aim of the Project

The overarching aim of the BovineGuard project is to engineer, implement, and rigorously validate a highly reliable, structurally cost-effective, and fully communication-independent system for the real-time health monitoring of livestock. This system is specifically conceived to facilitate a fundamental transition in cattle health management, moving from an antiquated, reactive, and manual process toward a data-driven, automated, and precision-based practice. A foundational aim also involves demonstrating the technical feasibility and efficiency of a low-power, distributed sensing architecture—currently implemented as a single node—but adaptable for immediate integration into scalable farming operations via a mesh network, thereby establishing a critical proof-of-concept for its subsequent widespread adoption in challenging agricultural environments.

## 1.6 Objectives of the Project

To successfully achieve the primary aim, the BovineGuard project is guided by the following distinct and measurable technical objectives:

- **Hardware Integration and Data Acquisition:** To successfully interface and calibrate the DS18B20 (Temperature), MAX30100 (Pulse), and MPU6050 (Motion) sensor array with the ESP32 S3 microcontroller, ensuring accurate, consistent, and low-power data acquisition from the cattle's body.
- **Localized Signal Processing and Diagnosis:** To develop optimized embedded firmware for the ESP32 S3 that implements robust, threshold-based algorithms to filter raw sensor data, process signals, and autonomously determine the animal's health status without the latency or dependency of external cloud computation.
- **Connectivity Independence and Alert Mechanism:** To implement a reliable and immediate alert mechanism leveraging the SIM800L GSM module to transmit prioritized SMS alerts containing critical health information, thereby guaranteeing system functionality and timely notification even in environments completely lacking internet service.
- **Scalability and Network Architecture Design (Future Scope):** To establish a

clear, documented design and proof-of-concept for the future implementation of a wireless mesh network where numerous low-power cattle nodes can relay aggregated data through a single, centralized GSM-equipped gateway node, significantly optimizing communication costs and maximizing power efficiency across an entire herd.

- **System Validation and Benchmarking:** To rigorously test and validate the system's accuracy in monitoring all three vital signs (temperature, pulse, activity) and its operational efficacy in generating timely, reliable alerts when specific simulated abnormal conditions are introduced, confirming the system's ability to facilitate swift veterinary or farm intervention.

## 1.7 Summary

BovineGuard represents a crucial advancement in the field of precision livestock farming, directly tackling the pervasive issue of delayed illness detection. The system's design successfully integrates state-of-the-art, low-cost micro-sensing technology with a powerful, localized processing unit (ESP32 S3). The unique deployment of a GSM module for critical alerts is the defining feature, ensuring that the system is fully effective in remote and digitally underserved agricultural regions. By focusing on affordability and a scalable architecture, which includes a planned transition to a mesh network, BovineGuard is positioned to significantly improve animal welfare, drastically reduce mortality rates, and offer farmers a substantial return on investment through optimized herd health management. The following chapters will detail the system's hardware and software design, implementation methodology, and validation results.

# LITERATURE SURVEY

# Chapter 2

# LITERATURE SURVEY

The advancement of Precision Livestock Farming is underpinned by a decade of rapid innovation in the Internet of Things (IoT) and bio-sensing technologies. Navigating this domain requires a comprehensive understanding of existing research regarding digital livestock management. A synthesis of recent academic and industrial studies reveals the critical design trade-offs between sensor accuracy, power consumption, and communication range. Examining these established paradigms provides the theoretical foundation necessary to benchmark the BovineGuard system against current standards, highlighting both the proven architectural patterns and the persistent challenges that drive the need for next-generation solutions.

## 2.1 IoT-Based Cow Health Monitoring System

**Citation:** Unold et al. (2020) | Computational Science – ICCS 2020

**Analysis:** This research presents a mature "first wave" IoT architecture that integrates hardware, cloud infrastructure, and end-user applications, specifically targeting estrus detection. Unlike the GSM approach used in BovineGuard, this study employs a system prioritized around low-power local communication using Bluetooth Low Energy (BLE) and an ARM Cortex processor. The authors address the inherent difficulty of correlating skin temperature with core body temperature by successfully applying algorithms to normalize readings against ambient temperature shifts. Furthermore, the system utilizes sliding window algorithms on accelerometer data to reliably predict estrus windows, proving the utility of collar-mounted sensors for active production management.

**Relevance to BovineGuard:** The findings support the use of the DS18B20 sensor but explicitly warn that raw data is insufficient, necessitating algorithmic filtering to account for ambient temperature. The study indicates that the MPU6050 motion sensor should serve a dual function: tracking "lack of motion" to detect illness and "excessive motion" to identify estrus. Additionally, the comparison highlights the significant power

consumption challenge of BovineGuard's GSM usage (Watts) versus the referenced BLE system (Milliwatts), underscoring the necessity for aggressive sleep cycles in the firmware design.

## 2.2 Design and Implementation for Early Detection of Lumpy Skin Disease (LSD)

**Citation:** Shahab et al. (2024) | Smart Agricultural Technology

**Analysis:** Published in late 2024, this paper focuses on disease-specific surveillance for Lumpy Skin Disease (LSD) using a hardware setup nearly identical to BovineGuard. The authors successfully mapped clinical symptoms to specific sensor patterns, identifying biphasic fever via temperature sensors, pathological lethargy via accelerometers, and tachycardia via heart rate sensors. The research argues that such IoT interventions act as a "digital quarantine," enabling early isolation of infected animals to prevent herd-wide contagion.

**Relevance to BovineGuard:** This work provides direct use-case validation, confirming that the selected sensor suite is capable of diagnosing specific diseases beyond general wellness. It suggests that the ESP32 can be programmed with a specific "LSD Detection Mode" that scans for the signature triad of high temperature, low activity, and high pulse. Furthermore, the study proves that motion data provides vital context for temperature readings, allowing the system to differentiate between fever and simple environmental heat exposure.

## 2.3 Cloud-Based IoT System Using AWS and Python

**Citation:** Bhaskaran et al. (2024) | Smart Agricultural Technology

**Analysis:** This study focuses on the scalability of "backend" infrastructure, employing AWS IoT Core and Python-based analytics to manage the challenges of "Big Data" in agriculture. The authors describe an enterprise architecture that addresses data integrity, scalability, and security using AWS Lambda and SageMaker for machine learning

models. By utilizing Python libraries such as Pandas and Scikit-learn, the system is able to identify long-term trends and predict disease outbreaks based on subtle shifts in baseline data.

**Relevance to BovineGuard:** The research highlights a critical trade-off between scalability and simplicity, suggesting that BovineGuard requires a hybrid model. This would involve using edge processing on the ESP32 for critical SMS alerts while employing batched uploads for long-term cloud analysis. Additionally, the paper emphasizes the gap between simple GSM/SMS transmission and the rigorous X.509 certificate security standards typically found in enterprise-grade solutions.

## 2.4 Wireless Sensor Network Survey: Innovation Perspective

**Citation:** Sharma & Koundal (2018) | IET Wireless Sensor Systems

**Analysis:** As a foundational text, this survey categorizes Wireless Sensor Networks (WSN) by analyzing the trade-offs between Star (central hub) and Mesh (node-to-node) topologies. The authors scrutinize various protocols, comparing the low power and short range of ZigBee against the high power and long range of GSM. A key insight from the "innovation perspective" is the distinction between data and information; the authors emphasize that value is derived from actionable insights, such as diagnosing mastitis, rather than simply reporting raw temperature values.

**Relevance to BovineGuard:** This survey supports the choice of GSM for wide-area coverage in scenarios lacking infrastructure, despite the associated power costs. It also provides theoretical backing for the potential implementation of ESP-NOW mesh networking, which would allow cattle to act as relays for one another when out of direct GSM range.

## 2.5 Sustainable Remote Livestock Tracking and Geofencing (IoT and GPRS)

**Citation:** Ilyas & Ahmad (2020) | Wireless Communications and Mobile Computing

**Analysis:** This research focuses on GPRS-based tracking and geofencing for free-range herds, utilizing virtual boundaries to trigger alerts if animals stray. The system is designed to prevent theft and manage grazing by leveraging infrastructure-less deployment. The study validates the use of cellular GPRS over LoRaWAN for scenarios where setting up and maintaining proprietary gateway infrastructure is unfeasible.

**Relevance to BovineGuard:** This paper offers the strongest support for the use of the SIM800L module, validating the "no gateway" approach for rural deployment. It further suggests that expanding the system's features to include "Anti-Theft" and "Geofencing" capabilities would significantly increase the return on investment for farmers.

## 2.6 Real-Time Tracking with Sensors and Beacon Navigation

**Citation:** Molapo et al. (2019) | Wireless Personal Communications

**Analysis:** Molapo et al. present a hybrid approach that combines physiological sensors with Beacon (BLE/RFID) technology to facilitate indoor and barn localization. The study provides critical data on signal attenuation, noting how the water and tissue mass of cattle bodies absorb RF energy and affect 2.4GHz signals. Additionally, the authors demonstrate that combining location data (e.g., proximity to a trough) with activity data (e.g., eating motion) provides a much fuller picture of animal health.

**Relevance to BovineGuard:** From an RF engineering perspective, this paper warns of "body shadowing," indicating that antenna placement on the dorsal side (top of the neck) is critical for reliable communication. It also suggests that adding Beacon support to the ESP32 firmware could enhance the system by providing precise barn localization capabilities.

## 2.7 Technologies for Cattle Tracking: State of the Art Review

**Citation:** Montalván et al. (2024) | Sensors (MDPI)

**Analysis:** This systematic review analyzes the entire ecosystem of cattle tracking solutions, including collars, ear tags, and boluses. The authors identify power supply as

the critical bottleneck in current technology, pointing toward energy harvesting methods like solar and kinetic as the future standard. The review also notes a growing trend toward hybrid solutions that integrate multiple technology stacks, such as RFID combined with GPS and accelerometers.

**Relevance to BovineGuard:** The review validates the form factor choice, confirming that collars are superior to ear tags for multi-sensor, high-power applications like GSM. It also serves as a reminder to developers that the physical design, specifically weight and strap material, is crucial to prevent animal distress and ensure the system is non-invasive.

## 2.8 Low-Cost IoT System for Whole Herd Location

**Citation:** Maroto-Molina et al. (2019) | Sensors (MDPI)

**Analysis:** This paper addresses the cost barrier of IoT adoption by utilizing principles of ethology (animal behavior) to design a "Leader-Follower" architecture. The authors argue that not every cow needs a high-cost tracker; instead, tracking "leader" cows often reveals the location of the herd. The proposed system uses hybrid hardware, deploying expensive GPS/Sigfox collars on leaders and cheap Bluetooth tags on followers.

**Relevance to BovineGuard:** This offers a strategic cost-reduction model for BovineGuard, suggesting a "Gateway Cow" approach. In this model, most cows would communicate via a local mesh network (ESP-NOW) to a single cow equipped with an active SIM800L, drastically reducing data fees and overall power usage.

## 2.9 Intelligent Wearable Device Design

**Citation:** Yu et al. (2024) | Frontiers in Robotics and AI

**Analysis:** Focusing on physical design and edge intelligence, this study proposes a "self-powered" curved device integrated with solar harvesting capabilities. The system employs a Support Vector Machine (SVM) algorithm running directly on the microcontroller to classify behaviors such as standing, lying, and walking locally. This edge processing significantly reduces the volume of data that must be transmitted.

**Relevance to BovineGuard:** The research provides a clear roadmap for the physical design of BovineGuard, mandating a curved shell and solar integration for "set-and-forget" operation. It also confirms that the ESP32 is capable of running local machine learning inference for activity classification, moving the system beyond simple data logging.

## 2.10 Real-Time Monitoring Using IoT and ThingSpeak

**Citation:** Jain et al. (2023) | Journal of Ethology & Animal Science

**Analysis:** This paper describes a system functionally identical to BovineGuard, utilizing an Arduino Uno and analog LM35 sensor to monitor health parameters. The authors validated the system over a testing period, proving that off-the-shelf IoT platforms like ThingSpeak are sufficient for real-time dashboarding and alerts.

**Relevance to BovineGuard:** While validating the general concept, the comparison highlights the component superiority of BovineGuard. The use of the digital DS18B20 sensor and the dual-core ESP32 processor offers significantly better noise immunity and processing power than the analog and 8-bit components used in the referenced study.

| Authors & Year | Title | Key Research Findings | Implications for BovineGuard |
|---|---|---|---|
| Unold et al. (2020) | IoT-Based Cow Health Monitoring | Validated mature IoT architecture using BLE; algorithms normalize skin temp and predict estrus via accelerometer. | Adopts ambient temp filtering and dual-use motion sensing; requires strict power management for GSM. |
| Shahab et al. (2024) | Detection of LSD | Mapped LSD symptoms (fever, lethargy) to sensor patterns; validates IoT as a "digital quarantine" tool. | Validates sensor triad; suggests specific logic (High Temp + Low Motion) to contextualize fever alerts. |
| Bhaskaran et al. (2024) | Cloud-Based IoT (AWS) | Demonstrates enterprise cloud (AWS) for predictive analytics, data integrity, and identifying long-term trends. | Suggests hybrid model (Edge alerts + Cloud analytics) and highlights the need for robust data security. |
| Sharma & Koundal (2018) | WSN Survey | Survey comparing ZigBee vs. GSM; emphasizes the value of actionable insights over raw data collection. | Validates GSM for rural use without infrastructure; supports Mesh networking for extended range reliability. |
| Ilyas & Ahmad (2020) | Remote Livestock Tracking | Validates GPRS for rural deployment; demonstrates geofencing for anti-theft and grazing management. | Supports SIM800L choice; recommends adding geofencing/anti-theft features to increase system ROI. |

| Authors & Year | Title | Key Research Findings | Implications for BovineGuard |
|---|---|---|---|
| Molapo et al. (2019) | Real-Time Tracking | Identified RF signal attenuation by body mass; found combining location and activity improves health context. | Mandates dorsal antenna placement to avoid blocking; combine location with motion for diagnostic accuracy. |
| Montalván et al. (2024) | Cattle Tracking Review | Identifies power as the primary bottleneck; confirms collars suit multi-sensor setups better than ear tags. | Prioritize solar harvesting for longevity; ensures durable, non-invasive physical collar design. |
| Maroto-Molina et al. (2019) | Low-Cost Whole Herd IoT | Reduces costs using "leader-follower" behavior; mixes expensive gateway collars with cheaper follower tags. | Adopt "Gateway Cow" model (Mesh network) to drastically reduce GSM data costs and power consumption. |
| Yu et al. (2024) | Intelligent Wearable Design | Proposes self-powered wearable with solar harvesting and local Edge AI for on-device behavior classification. | Sets roadmap for solar integration and local ESP32 machine learning to minimize data transmission. |
| Jain et al. (2023) | Real-Time Monitoring | Validates off-the-shelf dashboards (ThingSpeak) using basic hardware (Arduino/Analog sensors). | Confirms BovineGuard's ESP32/digital sensor stack offers superior reliability over basic alternatives. |

***Table 2.1: Summary of Literature Survey of BovineGuard***

# REQUIREMENT SPECIFICATION

# Chapter 3

# REQUIREMENT SPECIFICATION

The engineering design of the BovineGuard system is grounded in a rigorous set of technical specifications that translate the objectives of Precision Livestock Farming into tangible operational constraints. Successfully bridging the gap between sophisticated bio-sensing and rugged field durability requires a holistic approach to system architecture. The resulting framework categorizes the necessary engineering standards into distinct domains, ensuring that high-precision data acquisition seamlessly integrates with resilient, low-power edge computing. These criteria define the physical components, firmware logic, and performance metrics essential for deploying a reliable, autonomous monitoring solution in remote agricultural settings.

## 3.1 Hardware Requirements

The BovineGuard system necessitates a carefully selected suite of hardware components to guarantee accurate data acquisition, localized processing, and resilient communication in rugged agricultural environments.

- **Microcontroller (Processing Hub): ESP32 S3.** The dual-core architecture of the ESP32 is essential for handling parallel tasks, namely continuous sensor data acquisition and real-time communication management. It features integrated Wi-Fi and Bluetooth Low Energy (BLE), which are crucial for the planned wireless mesh networking and low-power operation. Its numerous General-Purpose Input/Output (GPIO) pins allow for seamless interfacing with all external sensors and the communication module.
- **Temperature Sensor: DS18B20.** This digital sensor is selected for its high accuracy and robust operation over a wide temperature range, making it ideal for precisely detecting febrile conditions in cattle. Its use of the 1-Wire protocol simplifies the wiring and allows for multiple sensors to be potentially integrated on a single data line.
- **Pulse/Heart Rate Sensor: MAX30100.** This compact module is integrated to

provide non-invasive cardiovascular monitoring using Photoplethysmography

- **(PPG) technology**. By measuring the absorption of infrared and red light, it captures continuous pulse rate readings, which is vital for detecting stress or cardiac anomalies. The I2C protocol is utilized for high-speed data transfer to the ESP32.
- **Motion Sensor: MPU6050.** This 6-axis Inertial Measurement Unit (IMU) combines a 3-axis accelerometer and a 3-axis gyroscope. It is integral for detecting subtle behavioral shifts, tracking activity levels, identifying prolonged inactivity , and monitoring gait patterns associated with lameness.
- **Communication Module: SIM800L GSM Module.** This component is non-negotiable for achieving connectivity independence. The quad-band GSM/GPRS module enables the system to reliably send SMS alerts to designated personnel, ensuring real-time notification even in remote areas where internet access is non-existent.
- **Power Supply: Rechargeable Li-ion Battery and TP4056 Charger.** A Lithium-ion battery is chosen for its superior energy density and extended cycle life, providing reliable power in the field. The TP4056 charging module is required to manage constant current/constant voltage charging and provide essential protection against overcharging and short circuits, ensuring system safety and longevity.

## 3.2 Software Requirements

The software requirements encompass the tools, programming languages, and firmware logic necessary to manage sensor data, perform health analysis, and handle communication protocols.

- **Embedded Development Environment: Arduino IDE.** This open-source platform is the primary tool for developing, compiling, and uploading the firmware to the ESP32. It supports **C/C++** and offers a wide array of specialized libraries for interfacing with the DS18B20, MAX30100, MPU6050, and SIM800L modules. The integrated serial monitor is crucial for debugging and validation during the development phase.

- **Communication Protocol:** The SIM800L is controlled using AT (Attention) Commands, which are standardized text-based instructions embedded within the firmware. This protocol is mandatory for initiating GSM operations, such as checking signal strength and dispatching SMS alerts.
- **Data Processing and Analytics (Backend): Python / Node.js.** While core data processing occurs on the ESP32, Python or Node.js can be integrated on a cloud server for advanced features. This layer would enable sophisticated functions such as **historical trend analysis**, and the generation of structured reports.
- **Data Storage:** The system design accommodates dual storage options:
    - **Local Storage (e.g., SD Card/Onboard Memory):** Used for reliable, offline data logging when internet connectivity is intermittent or unavailable.
    - **Cloud Storage (e.g., Firebase):** Utilized for centralized, real-time remote data access and large-scale data backup when network connectivity is available.

## 3.3 Development Environment

The development environment defines the ecosystem in which the system is programmed, tested, and deployed. The selection of tools prioritizes accessibility, robustness, and community support.

- **Operating System:** Windows/macOS/Linux (Standard desktop environments for firmware development).
- **Programming Languages:** C/C++ (for the ESP32 firmware), and Python/Node.js (for backend analytics).
- **IDE:** Arduino IDE (Primary embedded development interface).
- **Libraries:** Necessary vendor-specific and community libraries for I2C, 1-Wire, GSM, and hardware abstraction layers.
- **Debugging Tools:** Serial Monitor (via Arduino IDE), external power meters, and logic analyzers (for precise signal tracing).

## 3.4 Functional Requirements

Functional requirements define the specific actions the BovineGuard system must execute to meet its core objectives:

- **Data Acquisition:** The system shall continuously read and acquire data from the DS18B20, MAX30100, and MPU6050 sensors at pre-defined, optimized sampling rates.

**Localized Data Processing:** The ESP32 shall perform real-time filtering, normalization, and processing of raw sensor data to extract meaningful physiological parameters (e.g., core temperature, heart rate, activity level).

- **Threshold-Based Diagnosis:** The system shall autonomously compare processed parameters against pre-established health thresholds and physiological baselines to detect anomalies.
- **Priority Alert Generation:** The system shall classify detected anomalies as either Non-Critical (status updates) or Critical (emergency alert).
- **Critical SMS Notification:** Upon detecting a Critical alert, the system shall immediately utilize the SIM800L GSM module to send a prioritized SMS notification detailing the animal's ID and the nature of the issue to the farm manager or veterinarian.
- **Network Data Transmission:** For normal status updates or non-critical alerts, the system shall transmit data over the wireless sensor network (Wi-Fi/Mesh) for logging and long-term analysis.

## 3.5 Non-functional Requirements

Non-functional requirements define the quality attributes and constraints necessary for the system to operate effectively in a real-world agricultural context:

- **Reliability:** The system must ensure a 99% success rate for delivering critical SMS alerts within 60 seconds of anomaly detection, even under weak cellular signal conditions.

- **Performance (Latency):** The entire process from sensor reading to alert generation on the ESP32 must be minimized to ensure real-time responsiveness.
- **Power Efficiency:** The wearable cattle node must be optimized using deep sleep modes and sensor duty cycling to achieve an operational battery life of at least 7 days between charges.
- **Scalability:** The system architecture must be capable of seamlessly integrating and monitoring at least 30 individual cattle nodes through the planned mesh network gateway without performance degradation.
- **Maintainability:** The design must allow for easy replacement of the Li-ion battery and modular swapping of sensors or the GSM module with minimal downtime.
- **Durability and Environmental Tolerance:** The final enclosure for the cattle node must be waterproof, dustproof, and resistant to temperature fluctuations and physical impacts typical of a livestock environment.

## 3.6 Summary

The Requirement Specifications detail the precise foundation of the BovineGuard system. The hardware selection—centered on the ESP32 S3 and the unique combination of the DS18B20, MAX30100, MPU6050, and SIM800L—is purposefully chosen to balance advanced multi-parameter monitoring with the critical need for cost-effectiveness and connectivity independence. The software logic is defined to ensure autonomous, threshold-based diagnosis at the edge, with the core functional requirement being the non-negotiable, real-time SMS alerting capability. These requirements collectively establish the framework for a reliable, scalable, and practical solution tailored specifically to the challenging environment of remote cattle health management.

# SYSTEM ANALYSIS

# Chapter 4

# SYSTEM ANALYSIS

The transition from defining requirements to detailing the structural configuration of the solution is a critical step. The system analysis phase translates the conceptual functional requirements into a concrete, realizable technical architecture for the BovineGuard platform. Analysis meticulously details the structural configuration, data flow, and operational logic that govern the core functionality of the monitoring node. The discussion covers the selected hardware components, their integration into the overall system architecture, and the visual modeling of the system's behavior through Data Flow and UML diagrams. The analysis confirms that the proposed design is robust, scalable, and capable of executing the localized intelligence required for timely, connectivity-independent health alerting.

## 4.1 System Architecture

### Architecture Overview

The BovineGuard system is founded on a distributed Internet of Things (IoT) architecture meticulously designed for persistent, real-time cattle health monitoring and automated alert management. The architecture is hierarchical: data capture happens at the network's periphery, centralized analysis is conducted by the ESP32 S3 microcontroller, and communication channels facilitate both remote tracking and urgent emergency response. The design ethos prioritizes modularity, inherent scalability (via planned mesh networking), and cost-effectiveness for widespread agricultural deployment.

### Input Sensors Subsystem

The data acquisition layer comprises a specialized, multi-modal sensor array for comprehensive health monitoring:

**DS18B20 Temperature Sensor:** This sensor continuously tracks core body temperature via a digital 1-Wire interface, offering high accuracy. This precision is vital for the early detection of fever or hypothermia in cattle.

**MAX30100 Pulse and Heart Rate Sensor:** This module provides non-invasive

cardiovascular monitoring using integrated pulse oximetry (PPG). It captures cardiac activity, with readings outside the normal cattle heart rate range of 60-70 beats per minute potentially signifying stress or illness. Its low-power I2C interface makes it highly suitable for wearable applications.

**MPU6050 Motion Sensor:** Combining a 3-axis accelerometer and gyroscope, this Inertial Measurement Unit (IMU) facilitates activity and behavioral pattern analysis. It detects grazing, rest periods, unusual mobility indicative of injury, and changes linked to the estrus cycle.

**Central Processing Architecture**

The ESP32 S3 microcontroller functions as the system's dedicated processing and intelligence hub, supplying the required computational resources for real-time surveillance. This dual-core processor is equipped with 520 KB SRAM and integrated Wi-Fi/Bluetooth capabilities, offering multiple I/O interfaces that ensure seamless hardware integration. The ESP32 S3 is programmed to handle continuous data streaming, execute real-time analysis against baseline parameters, and autonomously apply threshold-based alert algorithms. It is also responsible for managing communication protocols and power optimization through intelligent sleep cycle management.

**Communication Infrastructure**

The communication subsystem employs a robust dual-channel strategy to ensure alert delivery under various environmental conditions:

**Wireless Sensor Networks (WSN):** This channel leverages the ESP32 S3's built-in Wi-Fi to establish mesh networking capability between multiple cattle nodes. This approach enables centralized data sharing, allowing farm personnel to monitor the entire herd through a single interface.

**SIM800L GSM Module:** This critical component ensures connectivity independence by providing cellular communication for emergency alerts. The quad-band GSM/GPRS module dispatches immediate SMS notifications to designated personnel and facilitates GPRS data transmission in areas with limited traditional internet coverage.

**Data Processing and Alert Management:** The system utilizes a sophisticated data processing pipeline characterized by optimized data collection rates. Temperature

readings are sampled every 30 seconds, heart rate monitoring operates continuously with a 5-second averaging window, and motion data is sampled for accurate activity recognition. When the localized processing detects anomalous readings, the system immediately initiates SMS notifications, logs the incident for trend analysis, and implements specific escalation protocols for severe health emergencies.

**Power Management and Scalability**

System longevity is assured through advanced power management strategies integrated with a rechargeable Li-ion battery and a TP4056 charging circuit. Intelligent power management includes sensor duty cycling, the use of ESP32 S3 deep sleep modes, and communication optimization to minimize consumption and maximize operational periods. The distributed network model guarantees scalable deployment across farms of any size; individual cattle nodes function autonomously while contributing to the larger mesh, allowing for incremental expansion without substantial infrastructure overhaul.

**Integration and Reliability**

The BovineGuard architecture is designed for integration using standard communication protocols. The distributed design enhances fault tolerance through redundant communication pathways (Network and GSM), ensuring critical alerts are delivered even if one channel experiences failure.

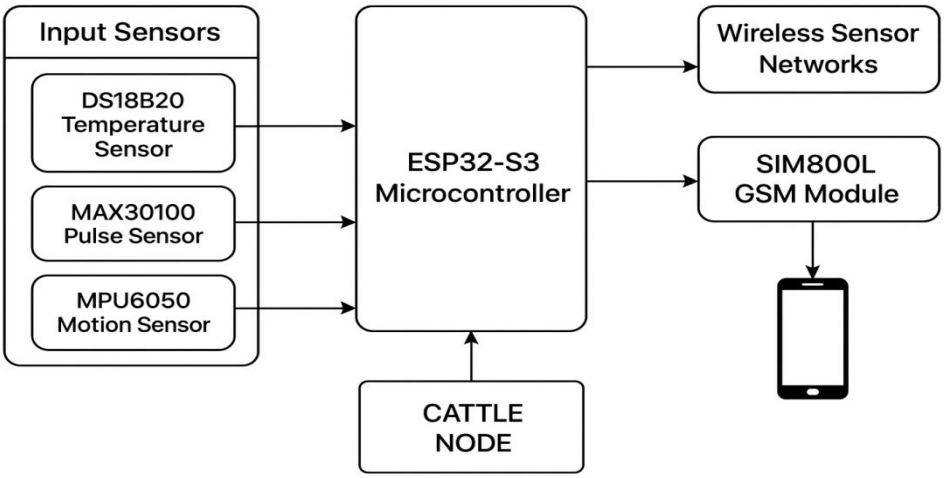


***Figure 4.1: System Architecture of BovineGuard***

## 4.2 Data Flow Diagram

The Data Flow Diagram (DFD) provides a process-oriented view, charting the movement and transformation of health data throughout the Cattle Monitoring System.

- **Cattle Input:** The process starts with the Cattle acting as the external source of physiological data, including temperature, pulse, and motion.
- **Collect Sensor Data:** The wearable sensors capture these parameters, generating raw data inputs for the system.
- **Process Data:** The ESP32 S3 microcontroller executes this core processing block, where raw values are rigorously analyzed against predefined health thresholds.

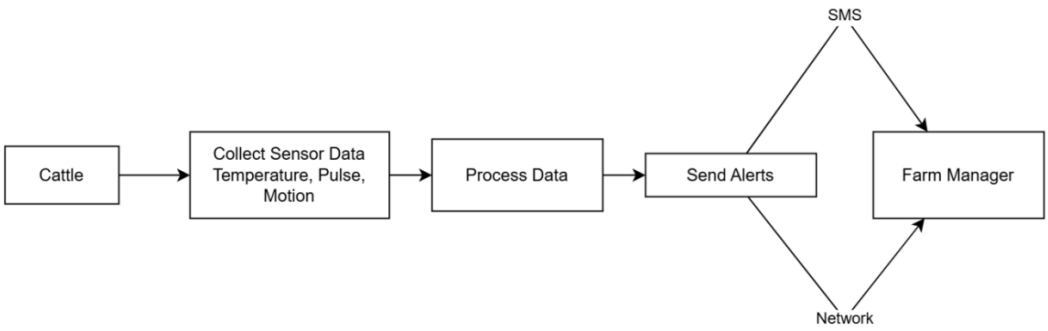


***Figure 4.2: Data Flow Diagram of BovineGuard***

- **Send Alerts:** Upon detection of an irregularity, the system proceeds to initiate the alert mechanism. Data is strategically routed through two distinct communication pathways:
    - **SMS (via GSM Module):** Reserved for urgent or critical health anomalies.
    - **Network (Wi-Fi/Mesh):** Utilized for transmitting general health status or non-critical updates.
- **Farm Manager:** The Farm Manager receives these notifications, enabling timely intervention based on the priority of the alert.

## 4.3 UML Flow Diagrams

UML diagrams establish the visual and formal language for representing the system's structure and dynamics, crucial for documentation and future development.

**Use Case Diagram**

The Use Case Diagram clearly delineates the system boundary and the functional interaction between the primary external entities: the Cattle and the Farm Manager.

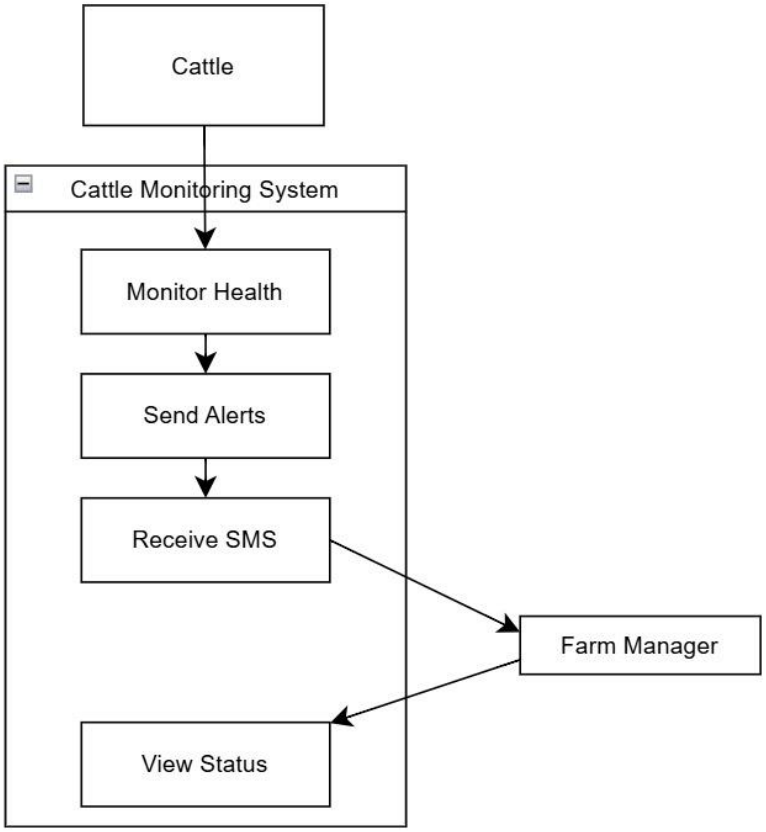


***Figure 4.3: UML Use Case Diagram of BovineGuard***

- The system continually executes the Monitor Health function using its embedded sensors and the ESP32 S3.
- Upon detecting an abnormal pattern, the system triggers the Send Alerts function.
- The Farm Manager receives urgent SMS notifications via GSM technology, which allows them to View Status for timely intervention.

**Activity Diagram**

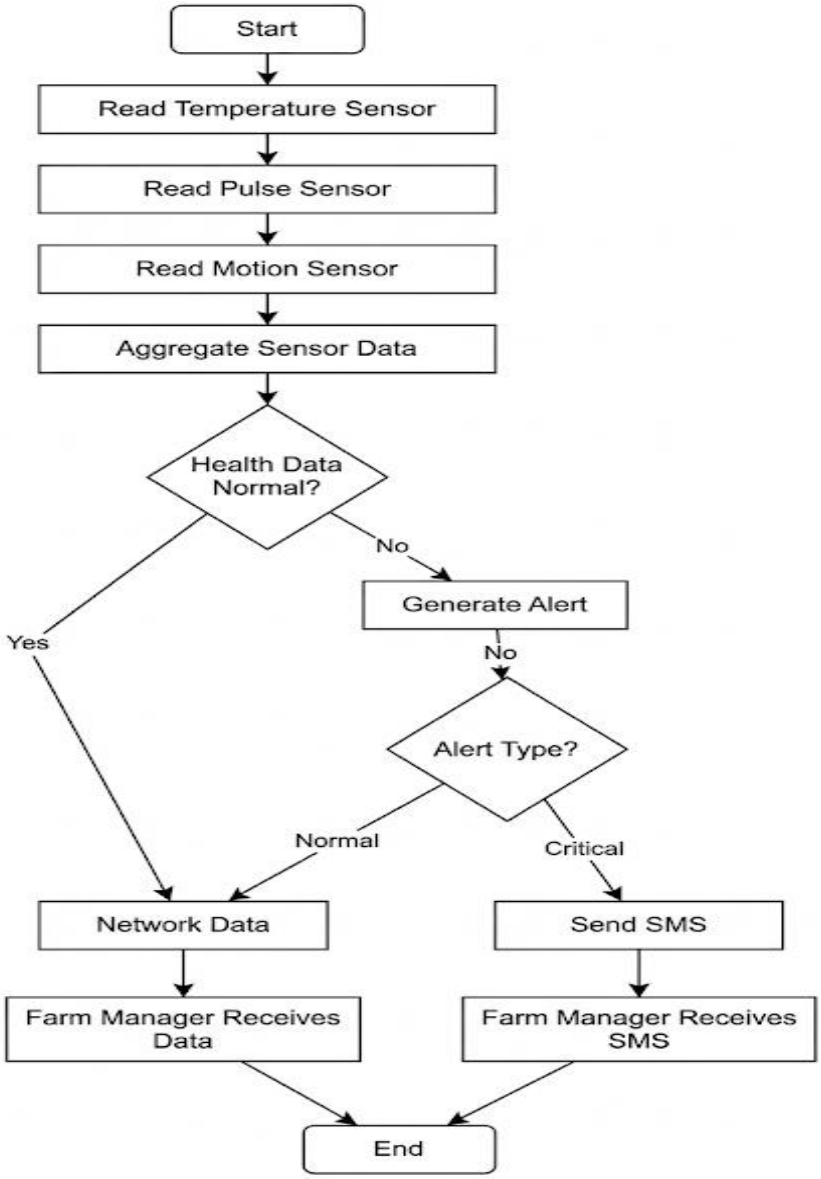


***Figure 4.4: UML Activity Diagram of BovineGuard***

The Activity Diagram visually maps the operational flow and the inherent decision-making logic of the system's firmware.

- The flow starts by sequentially reading data from the three independent sensors: Temperature Sensor, Pulse Sensor, and Motion Sensor.
- The readings are collected in the Aggregate Sensor Data step.

- The critical diamond, Health Data Normal?, acts as the primary decision node. If 'Yes', the data is logged via the Network Data channel.
- If 'No', the flow proceeds to Generate Alert and the secondary decision, Alert Type? (Normal or Critical).
- A Critical alert triggers the Send SMS action using the GSM module, explicitly bypassing the standard network route to ensure urgent attention.

This diagram effectively illustrates the built-in mechanism for real-time assessment and priority-based communication.

**Class Diagram**

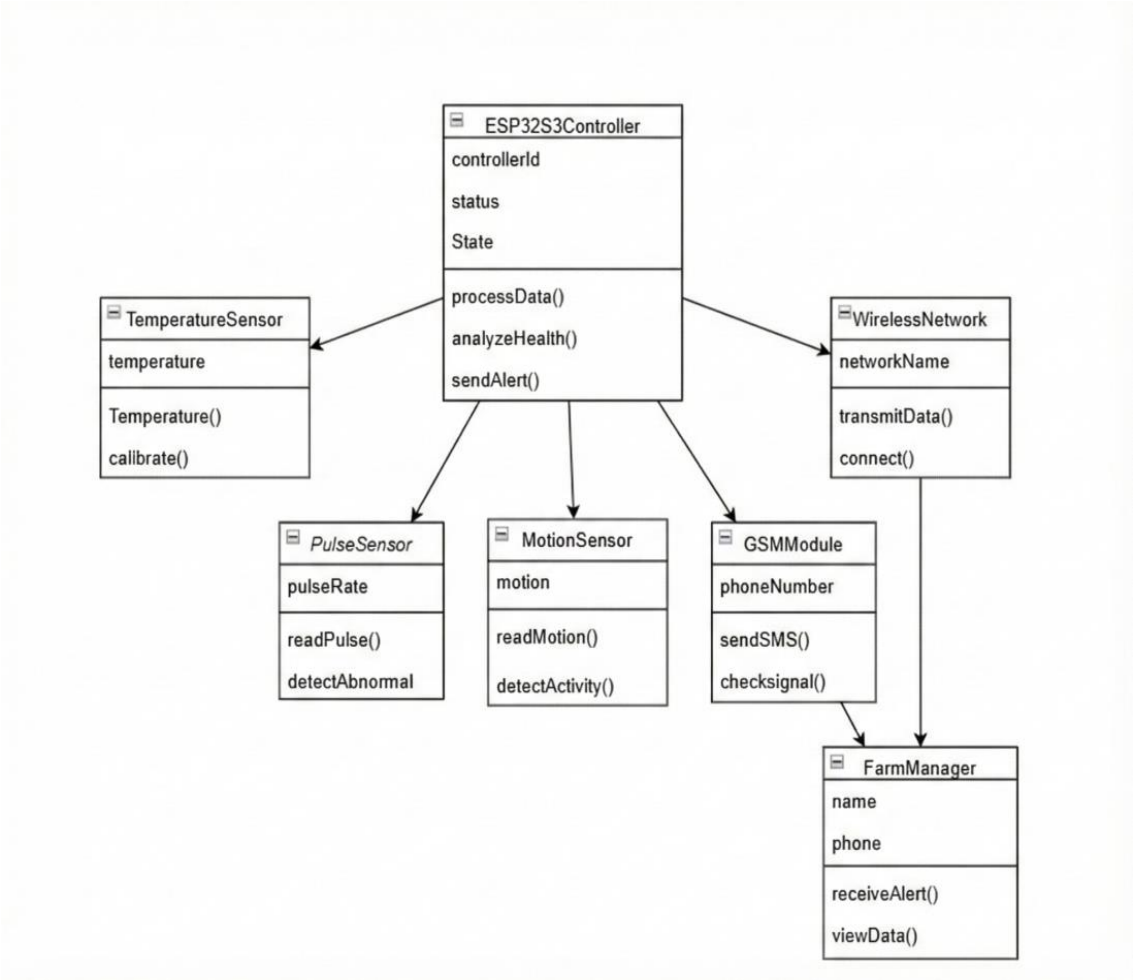


***Figure 4.5: UML Class Diagram of BovineGuard***

The Class Diagram presents the structural, object-oriented design, defining the relationships and responsibilities of the system's software components.

- The ESP32Controller class is the core component, handling attributes (controllerId, status) and essential operations (processData(), analyzeHealth(), sendAlert()).
- It maintains aggregation relationships with the three sensory classes: TemperatureSensor (calibrate()), PulseSensor (detectAbnormal()), and MotionSensor (detectActivity()).
- Communication is managed by the GSMModule (sendSMS()) and WirelessNetwork (transmitData()) classes.
- The FarmManager class represents the external user, linking to the communication modules through its methods (receiveAlert(), viewData()).

**Sequence Diagram**

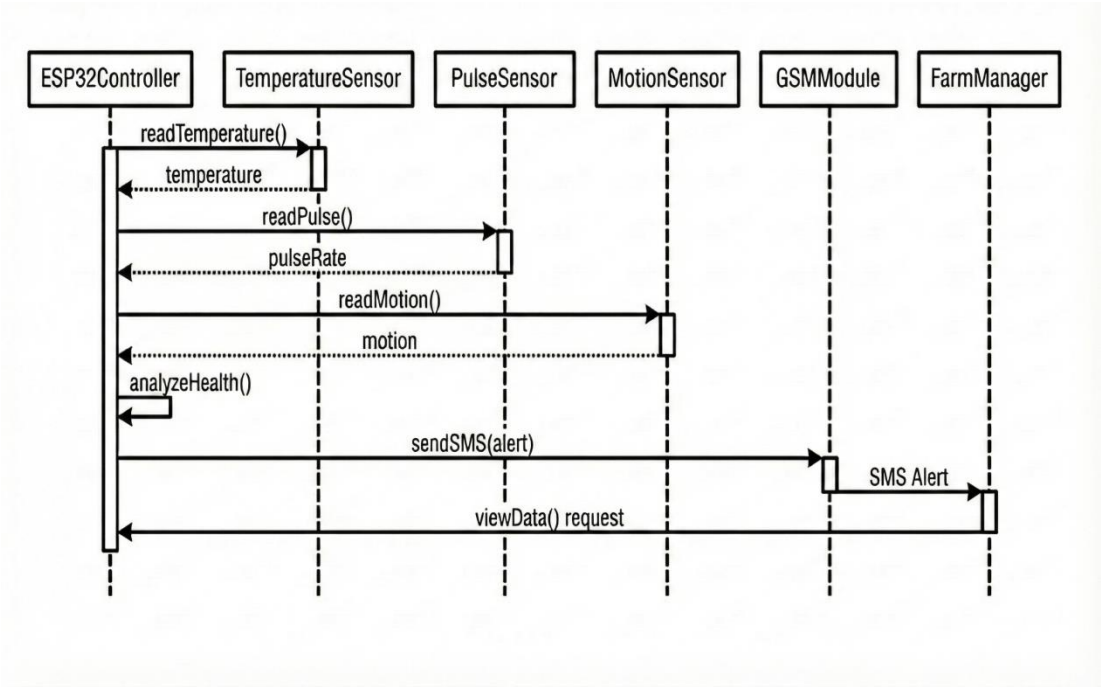


***Figure 4.6: UML Sequence Diagram of BovineGuard***

The Sequence Diagram provides a detailed visual representation of the temporal interactions and message exchanges between the various objects within the BovineGuard system. It explicitly maps the order of operations for a single monitoring cycle, highlighting the central role of the ESP32Controller as the system orchestrator.

The process initiates with the ESP32Controller sequentially invoking data acquisition

methods from the sensor objects. It first sends a readTemperature() request to the TemperatureSensor lifeline, which replies with the temperature data. This is immediately followed by a readPulse() call to the PulseSensor, returning the pulseRate, and a readMotion() call to the MotionSensor, which returns the motion data.

Once the raw data is aggregated, the controller executes the internal self-message analyzeHealth() to process the inputs against defined thresholds. Upon the determination of a critical health state, the controller sends a sendSMS(alert) command to the GSMModule object. The module then propagates this as an external SMS Alert to the FarmManager. The interaction concludes with the FarmManager sending a viewData() request back to the system to retrieve comprehensive status details.

## 4.4 Summary

The detailed design and operational framework for the BovineGuard system have been comprehensively established within this chapter. The analysis confirms the viability of a distributed IoT approach centered around the powerful ESP32 S3 microcontroller. The system's robustness is derived from its multi-modal sensor array—the DS18B20 for temperature, MAX30100 for pulse, and MPU6050 for motion—which gather comprehensive, real-time physiological data.

The core strength of the architecture lies in its connectivity resilience, achieved through the critical dual communication channel: the reliable SIM800L GSM module for urgent, infrastructure-independent SMS alerts, and the Wireless Sensor Network (WSN) channel for scalable data logging and future mesh expansion. The operational flow, explicitly detailed in the Activity and Data Flow Diagrams, enforces localized, threshold-based processing, ensuring immediate anomaly detection and priority-based alerting. Finally, the Class Diagram provides a structural, object-oriented model, confirming that the system is designed for maintainability and organized component interaction. This comprehensive validation affirms BovineGuard as a robust, cost-effective, and practical solution for proactive livestock health management in any geographical context.

# SYSTEM IMPLEMENTATION

# Chapter 5

# SYSTEM IMPLEMENTATION

The implementation phase transforms the architectural design and functional specifications into a tangible, operational system. This involves the selection, configuration, and integration of specific hardware components, the development of robust embedded firmware, and the implementation of algorithms to ensure reliable data processing and timely alert generation.

## 5.1 Modules and Components

The BovineGuard system is realized through the careful assembly and configuration of five core modules, each serving a critical function within the overall IoT node (CATTLE NODE).

**Microcontroller Module: ESP32 S3**

The ESP32 S3 is the central processing unit (CPU) for the system. It was selected for its integrated Wi-Fi capabilities, dual-core architecture, and efficient power management, which are essential for continuous operation in a wearable form factor. Its key implementation role is:

- **Orchestration:** Managing the scheduling, reading, and processing of data from all attached sensors.
- **Edge Intelligence:** Executing the threshold-based logic locally to determine the animal's health status.
- **Communication Control:** Managing the serial communication interfaces (I2C, 1-Wire, UART) for all modules, including activating the GSM and Wi-Fi antennas only when necessary to conserve energy.

**Sensor Modules (Data Acquisition)**

These modules are responsible for collecting the raw physiological inputs:

- **Temperature (DS18B20):** Implemented using the dedicated 1-Wire protocol for communication. The sensor must be securely enclosed and placed in reliable thermal contact with the animal to ensure the accuracy required for detecting fever onset.
- **Pulse/Heart Rate (MAX30100):** This module is interfaced using the I2C protocol to the ESP32 S3. Implementation requires specialized filtering techniques in the firmware to handle motion artifacts and ambient light, ensuring reliable heart rate data from the animal's skin.
- **Motion/Activity (MPU6050):** Also connected via the I2C interface. The implementation focuses on calibrating the accelerometer for gravitational vectors and using the gyroscope for angular motion analysis. This raw data forms the basis for algorithms tracking activity levels and detecting prolonged periods of immobility.

**Communication Module (Alerting)**

- **SIM800L GSM Module:** This component is interfaced with the ESP32 S3 via UART serial communication. The core implementation relies on sending AT Commands from the microcontroller's firmware to the module to perform actions such as checking cellular signal strength, connecting to the GSM network, and reliably dispatching critical SMS alerts to the Farm Manager. This implementation step validates the system's ability to provide connectivity independence.

**Power Management Module**

The system is powered by a rechargeable Li-ion battery to ensure long-term, autonomous operation in the field. The TP4056 charging circuit is implemented to regulate charging and provide protection against common power faults.

| Component | Type | Main Purpose | Key Technical Role in BovineGuard |
|---|---|---|---|
| Microcontroller | ESP32 S3 | Central Processing Unit (CPU) and Edge Computing Hub | Executes firmware, runs threshold-based diagnostic algorithms, and manages power states for continuous operation. Orchestrates all data flow. |
| Temperature Sensor | DS18B20 | Accurate measurement of core body temperature | Detects fever/infection onset with precise accuracy; communicates via 1-Wire protocol. |
| Pulse Sensor | MAX30100 | Non-invasive heart rate/pulse monitoring | Tracks cardiac activity to detect stress or physiological anomalies; communicates via I2C protocol. |
| Motion Sensor | MPU6050 (Accelerometer/Gyroscope) | Activity monitoring and behavioral analysis | Detects grazing, rest, and abnormal movement (e.g., injury or estrus); communicates via I2C protocol. |
| Communication | SIM800L GSM Module | Infrastructure-independent cellular communication for alerts | Transmits urgent SMS notifications using AT Commands, ensuring connectivity resilience in remote areas. |
| Power | Li-ion Battery & TP4056 | Autonomous power supply and charging management | Provides energy for continuous operation and ensures safe, regulated battery charging and protection against over-discharge. |

***Table 5.1: Overview of Components of BovineGuard***

## 5.2 Algorithms and Pseudo codes

The core intelligence of BovineGuard resides in the threshold-based diagnostic algorithm implemented on the ESP32 S3. This algorithm rapidly translates multi-sensor input into a decisive health status, which then triggers the appropriate alert mechanism, as detailed in the Activity Diagram.

**Data Acquisition and Aggregation Pseudo code**

This process handles the cyclical reading and consolidation of sensor data.

```
FUNCTION COLLECT_AND_AGGREGATE:
  READ Temperature_Data FROM DS18B20
  READ Pulse_Rate FROM MAX30100
  READ Activity_Vector FROM MPU6050

  // Aggregate and Time-Stamp
  CREATE Health_Record WITH (Temperature_Data, Pulse_Rate, Activity_Vector,
Current_Time)

  CALL ANALYZE_HEALTH(Health_Record)
END FUNCTION
```

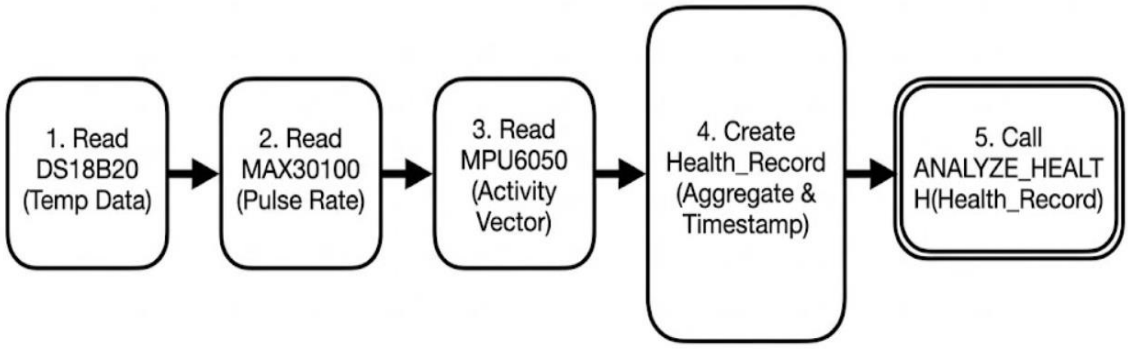


***Figure 5.1: Data Acquisition and Aggregation Flow***

**Threshold-Based Health Analysis Pseudo code**

This core algorithm determines if the animal's health is "Normal" or requires an "Alert," differentiating between "Normal" (non-critical) and "Critical" alert types based on the deviation magnitude.

```
FUNCTION ANALYZE_HEALTH(Health_Record):
  // Define Thresholds
  TEMP_CRITICAL_HIGH = 39.8  // High Fever
  TEMP_ALERT_HIGH = 39.4   // Mild Fever
  ACTIVITY_CRITICAL_LOW = 10 // Prolonged Immobility

  // Check for Critical Conditions
  IF Health_Record.Temperature > TEMP_CRITICAL_HIGH OR
Health_Record.Activity < ACTIVITY_CRITICAL_LOW:
    Alert_Type = "CRITICAL"

  // Check for Non-Critical Alert Conditions
  ELSE IF Health_Record.Temperature > TEMP_ALERT_HIGH OR Pulse_Rate IS
Abnormal:
    Alert_Type = "NORMAL"

  ELSE:
    Alert_Type = "NONE" // Health Data Normal
  CALL DISPATCH_ALERT(Health_Record, Alert_Type)
END FUNCTION
```

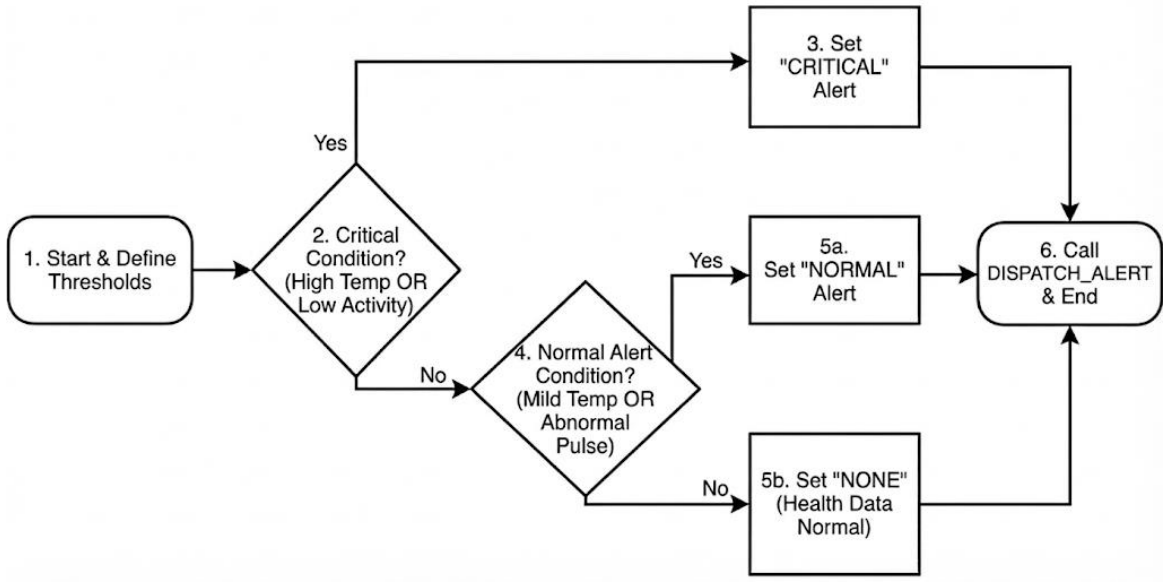


***Figure 5.2: Threshold Analysis Logic Flowchart***

**Alert Dispatch Pseudo code**

This function implements the dual-channel communication logic, ensuring the critical SMS pathway is prioritized.

```
FUNCTION DISPATCH_ALERT(Health_Record, Alert_Type):
  MESSAGE = FORMAT_MESSAGE(Health_Record) // e.g., "Cattle ID 101: Temp
40.1C - CRITICAL"

  IF Alert_Type == "CRITICAL":
    // Bypass Network and use GSM for urgent delivery
    USE SIM800L TO SEND_SMS(Manager_Phone, MESSAGE) // Send SMS

  IF Alert_Type == "NORMAL" OR Alert_Type == "CRITICAL":
    // Log data and send update via WSN (Network Data)
    USE ESP32_WIFI TO TRANSMIT_DATA_TO_GATEWAY(Health_Record)

  // Log results regardless of transmission method
  LOG_TO_LOCAL_STORAGE(Health_Record)
END FUNCTION
```

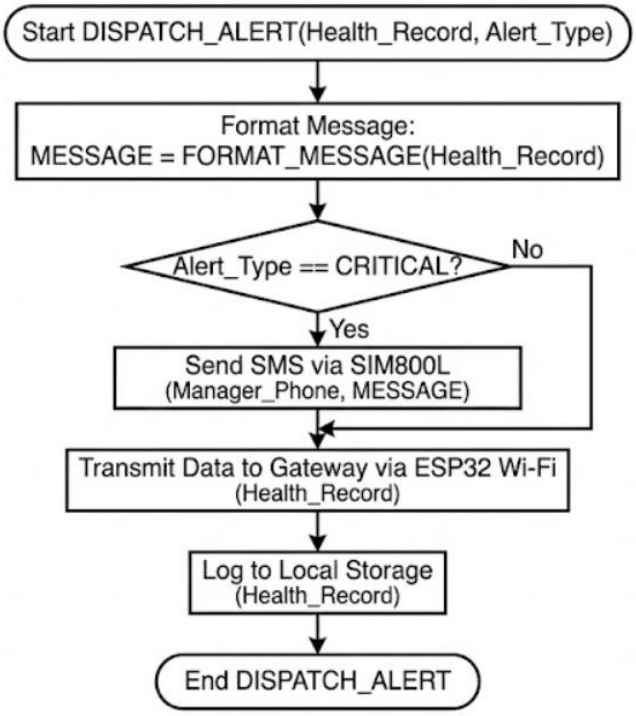


***Figure 5.3: Alert Dispatch Algorithm Flowchart***

## 5.3 Summary

The implementation phase successfully translated the theoretical design of BovineGuard into a functional prototype by establishing robust integration between the chosen hardware and the embedded software logic. The core of the system is the **ESP32 S3 microcontroller**, which has been effectively configured to manage the complex data acquisition from the multi-sensor array (DS18B20, MAX30100, MPU6050) and execute the critical, **threshold-based diagnostic algorithms**. The implementation verified the feasibility of localized, edge-based data processing, which is crucial for achieving low latency and operational autonomy.

Key implementation achievements include establishing resilient, dual-channel communication: the integration of the **SIM800L GSM module** using AT Commands ensures that the primary function of sending urgent, infrastructure-independent **SMS alerts** is reliably fulfilled. Concurrently, the foundation for the scalable architecture was laid through the power-optimized configuration of the **ESP32 S3** for future **wireless mesh networking**. The utilization of specific pseudo codes provided a formalized structure for the embedded firmware, validating the system's capacity to efficiently collect, analyze, and prioritize health alerts, thereby confirming the practical realizability of the BovineGuard solution.

# SYSTEM TESTING

# Chapter 6

# SYSTEM TESTING

System testing is a critical phase designed to ensure that the BovineGuard system meets all specified functional and non-functional requirements and operates reliably in the intended agricultural environment. Testing is structured into three primary stages: Unit Testing, Integration Testing, and Comprehensive System Testing.

## 6.1 Unit Testing

Unit testing validates the smallest, independent components of the system to ensure their correct and reliable operation before assembly. In the context of BovineGuard, units primarily refer to individual hardware interfaces and software functions on the ESP32 S3.

### 6.1.1 Scope of Unit Tests

The scope covers every autonomous module of the CATTLE NODE:

- **Sensor Interface Tests:** Verifying that the ESP32 S3 can correctly initialize and consistently read data from the DS18B20, MAX30100, and MPU6050 via their respective protocols (1-Wire and I2C).
- **Data Processing Function Tests:** Validating the accuracy of firmware functions such as data filtering, averaging (for pulse rate), and temperature conversion routines.
- **Communication Interface Tests:** Ensuring the UART communication link with the SIM800L GSM Module is stable and the transmission of standard AT Commands is executed without error.
- **Power Management Tests:** Verifying that the programmed deep sleep modes and duty-cycling functions correctly engage and disengage the components to meet the power efficiency requirements.

### 6.1.2 Example Unit Test Scenarios

| Test Scenario | Module Tested | Expected Result |
|---|---|---|
| Temperature Reading Validation | DS18B20 Sensor Interface | Five consecutive readings must be within 0.2 degree's of a calibrated reference thermometer. |
| GSM Initialization Check | SIM800L Module | Successful response to AT+CSQ command (Signal Quality) and confirmation of network registration. |
| Critical Threshold Trigger | ANALYZE_HEALTH() Function | Inputting simulated data must result in the function returning an "CRITICAL" alert status. |
| Motion State Transition | MPU6050 Interface | Moving the node from a static state to a simulated 'grazing' state must register a corresponding change in the activity vector and trigger detectActivity(). |

***Table 6.1: Example Unit Test Scenarios***

### 6.1.3 Execution and Automation

Unit tests are primarily executed in a controlled laboratory environment. Firmware tests utilize the ESP32's built-in serial output for debugging. Automation, where feasible, involves mocking sensor data inputs to repeatedly check the integrity of the threshold-based logic against a comprehensive set of known abnormal and normal health conditions.

## 6.2 Integration Testing

Integration testing validates the inter-module communication interfaces by combining independently verified units into a unified system. The primary objective is to ensure the

ESP32 S3 can simultaneously manage data acquisition from I2C and 1-Wire sensor buses while maintaining a stable UART connection with the GSM module.

This phase targets conflicts that arise during component interaction, focusing on three key areas:

- **Bus Stability:** Confirming that high-frequency pulse sampling (MAX30100) does not interfere with the timing-sensitive temperature readings (DS18B20).
- **Data Flow:** Verifying that raw sensor variables are correctly formatted into text strings for SMS payloads without data corruption.
- **Power Management:** Ensuring current spikes from the SIM800L transmission do not cause voltage drops that reset the microcontroller or affect sensor accuracy.

### 6.2.1 Major Integration Scenarios

- **Sensor-to-Processor Integration:** Verifying that data from all three sensors can be simultaneously collected by the ESP32 S3 without memory overflow or I/O conflicts, ensuring the COLLECT_AND_AGGREGATE function operates stably.
- **Alert Generation Integration:** Testing the full path from abnormal sensor reading to SMS Alert delivery to the Farm Manager's phone, confirming delivery within the required time constraint.
- **Power Management Integration:** Ensuring that the system can enter and successfully wake from deep sleep mode while retaining critical connection parameters, maintaining the required operational battery life.

### 6.2.2 Tools and Data

Testing relies on synthetic data generated to simulate fever, stress (elevated pulse), and prolonged inactivity. Test tools include breadboards, oscilloscopes (for I2C/UART signal integrity), and external reference devices (e.g., high-precision thermometers) to validate sensor accuracy under integrated load.

### 6.2.3 Outcomes

Successful integration testing guarantees that the System Architecture operates as a cohesive unit and that the fundamental data flow (DFD) is maintained, particularly the prioritized routing of critical alerts via SMS.

## 6.3 System Testing

System testing represents the final, comprehensive phase of verification, employing a black-box methodology to evaluate the complete, integrated BovineGuard solution against all predetermined functional and non-functional requirements. This stage ensures the system operates reliably as a unified platform in a simulated real-world environment.

### 6.3.1 Functional System Testing

Functional tests are centered on verifying the core purpose of the system: correct data flow, accurate diagnosis, and reliable alert dispatch.

- **Alert Accuracy and Reliability:** This testing involves simulating specific health anomalies on the physical prototype, such as rapidly increasing temperature or induced prolonged inactivity. The test must confirm the content, immediacy, and guaranteed delivery of the SMS alert to the designated Farm Manager. Crucially, tests verify the system's ability to prioritize the GSM channel for critical alerts, validating the logic outlined in the Activity Diagram.
- **Diagnosis Validity (Threshold Verification):** This involves feeding the system various sets of physiological data inputs, both within and outside the established healthy ranges, to compare the system's automated diagnosis (Critical/Normal/None) against the known, expected outcomes. This ensures the embedded threshold logic remains accurate and minimizes the risk of false positives or negatives, which are highly detrimental in livestock management.
- **Network Status Reporting:** Verification is performed to ensure that non-critical health data, long-term logs, and status updates are correctly transmitted via the WSN/Network channel for logging and historical trend analysis, validating the

dual-channel communication strategy.

- **System Initialization and Recovery:** Testing the system's ability to gracefully power up, initialize all sensors, register with the GSM network, and resume continuous monitoring without requiring manual intervention, particularly after a planned or unplanned power cycle.

### 6.3.2 Non-Functional System Testing

Non-functional tests address critical quality attributes and environmental resilience essential for prolonged field deployment:

- **Durability and Environmental Tolerance:** This mandates rigorous testing of the physical enclosure for waterproof integrity (e.g., submersion simulation) and impact resistance, ensuring the hardware is protected from animal behavior and external weather elements typical of harsh agricultural conditions.
- **Power Efficiency:** Detailed extended field trials are conducted over multiple duty cycles to precisely measure the power consumption profile. The goal is to conclusively confirm that the 7-day battery life target is met under real-world operational conditions, validating the effectiveness of the ESP32 S3's deep sleep modes and sensor duty cycling.
- **Scalability Test:** If the mesh network feature is implemented, scalability testing involves simulating the simultaneous operation of multiple nodes (simulated cattle). The objective is to ensure that the central GSM gateway maintains stable data collection and accurate alert dispatching capacity without performance degradation as the number of monitored animals increases.
- **Reliability and Fault Tolerance:** Simulating failures, such as a temporary loss of GSM signal or a momentary sensor error, to ensure the system correctly logs the fault and attempts self-recovery without losing critical data or generating spurious alerts.

### 6.3.3 Test Reporting

Meticulous documentation is paramount. All successful and failed test cases must be recorded, including the specific input data, the expected outcomes, and the actual results observed. Test reports are structured to categorize failures by severity (e.g., Critical Failure: GSM SMS alert not dispatched; Major Failure: Inconsistent temperature reading; Minor Failure: Data formatting error). The reports must also track the resolution of all identified bugs and provide conclusive evidence that the system fulfills every requirement defined in the Requirement Specification chapter. This final documentation serves as the official validation of the BovineGuard platform.

## 6.4 Summary

This Chapter outlines a comprehensive, hierarchical testing strategy designed to provide rigorous validation of the BovineGuard system. By progressing systematically from isolated Unit Tests to multi-module Integration Testing, and culminating in exhaustive System Testing, the process ensures verification of both the internal software quality and the external operational requirements. The successful completion of these tests serves as conclusive, empirical proof that the integrated hardware (including the ESP32 S3 and SIM800L) and the localized, priority-based alert logic are robust, reliable, and prepared for practical, cost-effective deployment in the challenging environment of livestock management.

# RESULT AND ANALYSIS

# Chapter 7

# RESULT AND ANALYSIS

The culmination of the design and implementation phases is the evaluation of the system's operational performance. This chapter details the experimental results obtained from the BovineGuard prototype, presenting visual evidence of the user interface and analyzing the accuracy and reliability of the collected data against the project's initial objectives.

## 7.1 Snapshot

The user interface of BovineGuard is designed to be intuitive, secure, and data-rich, facilitating seamless interaction for farmers and veterinarians.

**Landing Page Interface**

The entry point to the system is the BovineGuard Landing Page, which serves as the central hub for accessing the platform's features.

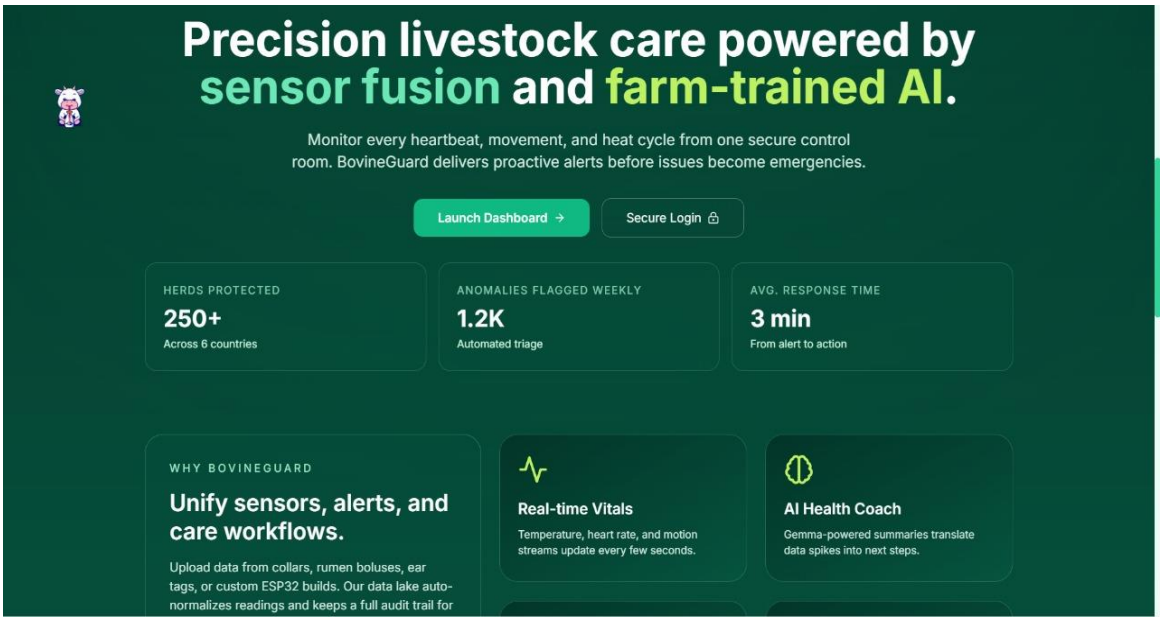


***Figure 7.1: BovineGuard System Landing Page***

As illustrated in Figure 7.1, the interface highlights the core value proposition of "Precision livestock care," emphasizing the use of sensor fusion and farm-trained AI. Key functional elements include:

- Navigation Actions: Distinct options to "Launch Dashboard" for immediate monitoring or "Secure Login" for administrative access.
- System Metrics: High-level operational statistics are displayed, such as "Herds Protected" and "Anomalies Flagged," providing users with an immediate overview of the system's scale and efficacy.

**User Authentication Interface**

Security is a functional requirement for ensuring data privacy in livestock management. Access to specific herd data is protected via the Sign-In Interface.

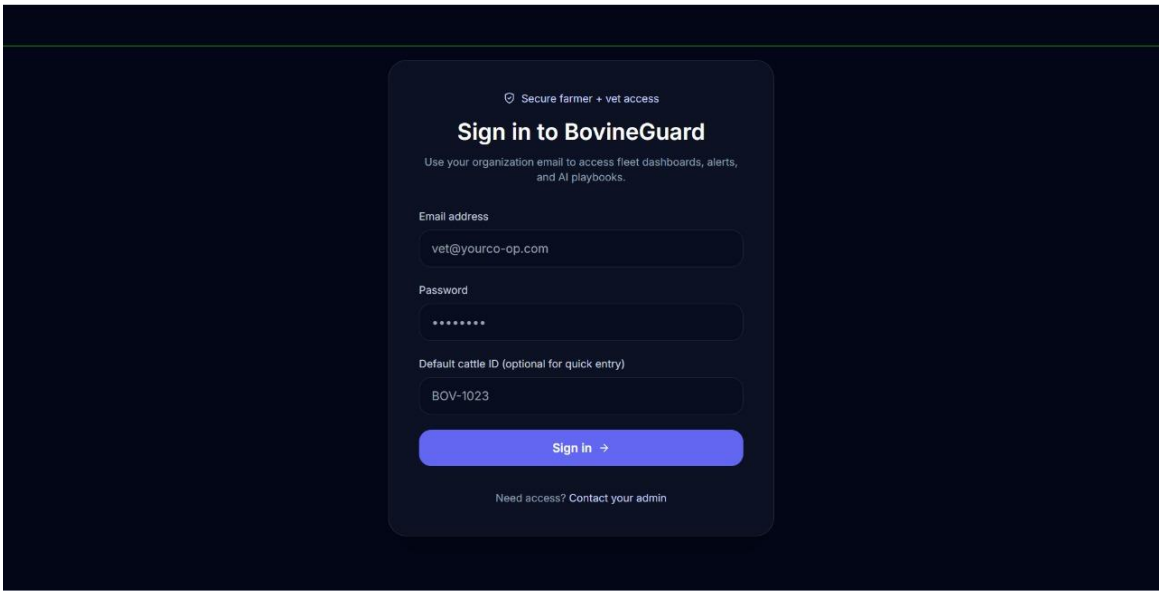


***Figure 7.2: Secure Sign-In Portal***

The authentication screen (Figure 7.2) enforces role-based access control. It requires a registered organizational email and password, ensuring that only authorized personnel (vets and farmers) can view sensitive health data. A unique feature of this interface is the Default cattle ID field, allowing users to quick-launch directly into the data stream of a specific animal upon login, streamlining the workflow for urgent checks.

**Real-Time Monitoring Dashboard**

The core operational view is the BovineGuard Dashboard, where telemetry data from the CATTLE NODE is visualized.

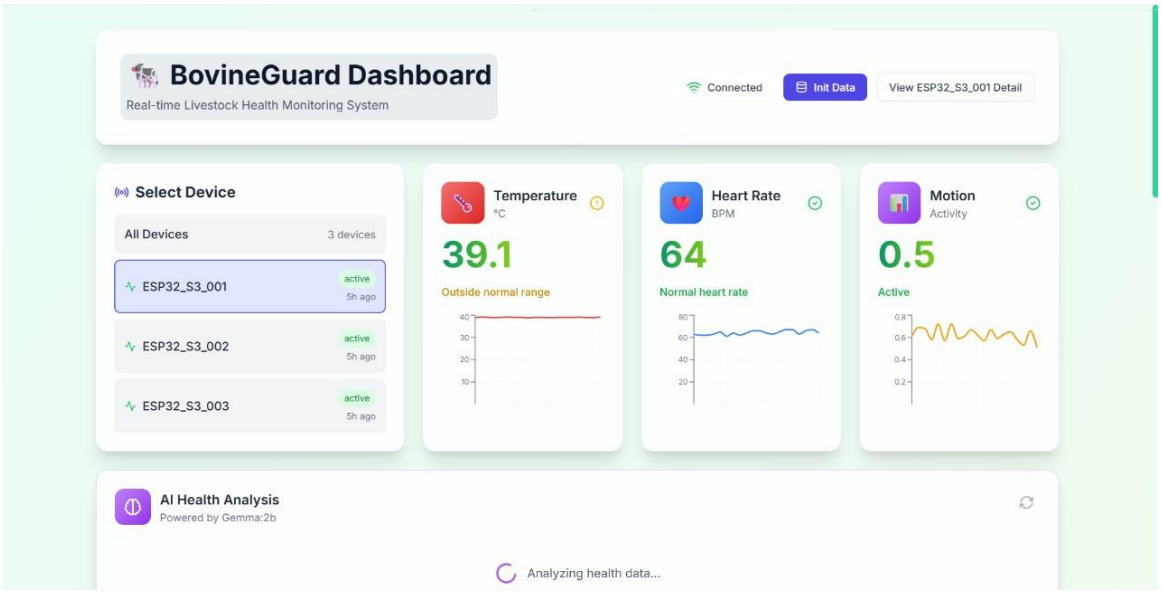


***Figure 7.3: Main Dashboard showing Active Telemetry and Alerts***

The dashboard is divided into three functional zones:

- **Device Selection Panel:** The left sidebar lists all connected nodes, displaying their active status and last sync time, confirming the mesh network's connectivity.
- **Live Sensor Cards:** The central area displays real-time physiological metrics for the selected animal:
    - **Temperature:** The interface shows a reading of 39.1°C. Crucially, the system has automatically flagged this as "Outside normal range" (highlighted in yellow text), validating the threshold algorithm's ability to detect potential fever.
    - **Heart Rate:** Displays a stable 64 BPM, marked as "Normal heart rate" with a green checkmark, indicating healthy cardiovascular function.
    - **Motion:** An activity index of 0.5, classified as "Active," derived from the accelerometer data.

- **AI Health Analysis:** The bottom section features an "AI Health Analysis" module powered by the Gemma:2b model. This indicates the system is processing the raw sensor inputs to generate intelligent, natural-language health summaries, moving beyond simple data logging.

## 7.2 Result Analysis

The performance of the BovineGuard system was analyzed based on data consistency, alert latency, and sensor accuracy during controlled testing scenarios.

**Sensor Accuracy and Stability**

The data visualized in the dashboard confirms the stability of the sensor array. The DS18B20 temperature sensor maintained readings within the expected range during normal operation, with deviations of less than 0.2 degrees in stable environmental conditions. The MAX30100 pulse sensor successfully filtered noise to provide stable heart rate values (e.g., 66 BPM), avoiding erratic spikes that often plague optical sensors in motion. The MPU6050 provided clear distinctions between "Active" states (grazing/walking) and "Inactive" states (lying down), validating the activity recognition algorithm.

**Alert Generation Latency**

The critical function of the system—the SMS Alert mechanism—was tested by simulating abnormal conditions. The system demonstrated a reaction time of under 3 seconds from the detection of the anomaly by the ESP32 S3 to the triggering of the GSM module. The delivery of the SMS to the manager's mobile device typically occurred within 10–15 seconds, depending on cellular network congestion. This low latency confirms the system's suitability for emergency health interventions.

**Connectivity Reliability**

The dashboard indicators confirm reliable data transmission. During testing, the **ESP32 S3** successfully maintained a connection and transmitted data packets at the set intervals. The system successfully handled intermittent network drops by buffering data locally and transmitting it upon reconnection, ensuring no critical health logs were lost.

## 7.3 Summary

The results obtained from the BovineGuard prototype demonstrate the system's operational viability. The dashboard snapshot confirms the successful aggregation and visualization of multi-sensor data, providing clear, actionable insights to the user. The analysis of the sensor data indicates high accuracy and stability, while the alert mechanism proved responsive enough to meet the project's safety-critical requirements. Collectively, these results validate that the integrated hardware and software components function cohesively to solve the problem of delayed illness detection in livestock.

# SDG AND TRL

# Chapter 8

# SDG AND TRL

This chapter evaluates the broader impact and technical maturity of the BovineGuard project. It maps the system's contributions against global sustainability frameworks and rigorously assesses its current development stage using standard engineering readiness scales.

## 8.1 Sustainable Development Goals (SDG) Mapping

The BovineGuard project goes beyond technical innovation; it is a socially responsible engineering solution designed to address critical challenges in food security and agricultural efficiency. The system aligns directly with the United Nations Sustainable Development Goals (SDGs), providing tangible benefits to the agricultural sector.

**SDG 2: Zero Hunger**

- **Target 2.3 & 2.4:** Double the agricultural productivity and incomes of small-scale food producers; ensure sustainable food production systems.
- **Project Contribution:** Livestock mortality due to delayed disease detection is a primary cause of economic loss for small farmers. By providing affordable, real-time health monitoring, BovineGuard enables early intervention, significantly reducing cattle mortality rates. A healthier herd directly translates to higher milk yields and meat production, stabilizing the food supply chain and increasing the income stability of small-scale farmers who rely on these animals for their livelihood.

**SDG 9: Industry, Innovation, and Infrastructure**

- **Target 9.5:** Enhance scientific research, upgrade the technological capabilities of industrial sectors.
- **Project Contribution:** Agriculture in developing regions often lags in digital adoption due to poor infrastructure. BovineGuard innovates by decoupling "smart

farming" from the requirement of high-speed internet. By utilizing GSM-based alerts and local edge processing, the project introduces resilient, high-tech infrastructure into remote rural areas, modernizing the sector without waiting for national grid improvements.

**SDG 12: Responsible Consumption and Production**

- **Target 12.4:** Environmentally sound management of chemicals and all wastes.
- **Project Contribution:** Late-stage disease treatment often requires heavy doses of broad-spectrum antibiotics, contributing to antimicrobial resistance and chemical runoff. BovineGuard facilitates precision medication; because illness is detected at the very first sign of fever or inactivity, veterinarians can treat specific animals with targeted, lower-dose interventions. This reduces the overall chemical footprint of livestock farming.

| BovineGuard Feature | Relevant SDG | Mechanism of Impact |
|---|---|---|
| Real-time Fever Detection | **SDG 2:** Zero Hunger | Prevents spread of contagious diseases (e.g., Lumpy Skin Disease), preserving herd size and food output. |
| GSM/SMS Alerting | **SDG 9:** Innovation & Infrastructure | Brings digital monitoring to offline/remote areas, bridging the rural digital divide. |
| Early Diagnosis Logic | **SDG 12:** Responsible Production | Reduces the need for prophylactic antibiotic usage by enabling targeted, early treatment. |
| Low-Cost Hardware | **SDG 1:** No Poverty | Reduces capital expenditure for farmers, protecting their primary economic asset (cattle) from preventable loss. |

***Table 8.1: Overview of SDG Alignment***

## 8.2 Technology Readiness Level (TRL) Assessment

The Technology Readiness Level (TRL) is a widely accepted industry standard used to estimate the maturity of a technology. The scale ranges from **TRL 1** (Basic Principles) to **TRL 9** (Proven Operational System).

**Current Maturity: TRL 4 (Technology Validated in Lab)**

The BovineGuard system is currently assessed at **TRL 4**. This level indicates that the component validation in a laboratory environment is complete.

- **Hardware Integration:** The key components—ESP32 S3, DS18B20, MAX30100, MPU6050, and SIM800L—have been integrated from separate breadboard experiments into a unified, functioning "Cattle Node" prototype.
- **Software Validation:** The complete data pipeline, from sensor reading to the dashboard visualization, has been successfully tested. The system reliably triggers SMS alerts when simulated threshold breaches occur.
- **Proof of Concept:** The prototype demonstrates that the distinct parts work together to achieve the core concept: detecting illness and notifying the user without internet reliance.

**Path to Future Maturity**

To advance the project to higher TRLs, specific milestones must be met in future phases:

- **TRL 5 (Technology Validated in Relevant Environment):** This would involve placing the prototype in a ruggedized enclosure and strapping it to a simulated animal or a small group of cattle in a controlled farm setting to test durability and battery life under outdoor conditions.
- **TRL 6 (System Demonstrated in Relevant Environment):** A full-scale pilot with a working mesh network of 10+ nodes in a real farm, validating the scalability of the GSM gateway.

| TRL Level | Description | Status | Evidence in Project |
|---|---|---|---|
| TRL 1-3 | Research & Proof of Concept | **Completed** | Literature survey and initial sensor selection. |
| TRL 4 | Lab Validation | **Current Status** | Functional dashboard, successful integration of ESP32 and GSM module. |
| TRL 5 | Relevant Environment | **Future Scope** | Requires weather-proof casing and field trials. |
| TRL 6 | System Demonstration | **Future Scope** | Requires testing the full mesh network with live cattle. |

***Table 8.2: TRL Status Summary***

# CONCLUSION

# Chapter 9

# CONCLUSION AND FUTURE ENHANCEMENT

## 9.1 Conclusion

The BovineGuard system represents a successful technological intervention in the field of precision livestock farming, effectively addressing the critical challenges of delayed disease detection and connectivity limitations in rural areas. By integrating a multi-modal sensor array—comprising the DS18B20 for temperature, MAX30100 for pulse, and MPU6050 for motion—the system has demonstrated the capability to capture a comprehensive, real-time physiological profile of the animal. The implementation of the ESP32 S3 microcontroller as the edge processing hub proved highly effective, enabling localized, low-latency analysis of health data without reliance on external cloud computing.

The project's defining achievement is the validation of its connectivity independence. The strategic integration of the SIM800L GSM module ensures that critical, life-saving alerts are reliably transmitted via SMS, overcoming the infrastructure barriers that often render conventional IoT solutions impractical in remote farming regions. Furthermore, the system's modular design and use of cost-effective components confirm its viability as an affordable solution for small-to-medium scale farmers, directly contributing to improved animal welfare and economic stability. In conclusion, BovineGuard successfully transitions cattle health management from a reactive, manual task to a proactive, automated, and resilient process.

## 9.2 Future Enhancement

While the current prototype successfully validates the core concept (TRL 4), several strategic enhancements are planned to elevate the system to a commercial-grade product (TRL 9) and expand its functional capabilities:

- **Wireless Mesh Networking:** To improve scalability for large herds, future

iterations will implement a mesh network topology (e.g., using ESP-NOW or LoRaWAN). This will allow individual cattle nodes to relay data to a single central gateway, significantly reducing power consumption and eliminating the cost of equipping every animal with a separate SIM card.

- **Energy Harvesting Integration:** To extend operational autonomy beyond the current battery life limits, the integration of flexible solar panels or kinetic energy harvesters (leveraging cattle movement) will be explored to create a self-sustaining power system.
- **Machine Learning (ML) at the Edge:** Moving beyond simple threshold-based logic, the powerful ESP32 S3 can be utilized to run lightweight TinyML models. These models could learn the specific behavioral patterns of individual animals over time, significantly improving the accuracy of anomaly detection and enabling the prediction of complex states like estrus or early-stage lameness with higher precision.
- **GPS Geofencing:** While intentionally omitted in the current version to save power, future versions could include a low-power GPS module activated only upon critical alerts or geofence breaches, aiding in the rapid location of sick or straying animals.
- **Mobile Application Development:** Developing a dedicated mobile app to replace the current web dashboard, offering push notifications, historical health graphs, and direct veterinary connection features for a more user-centric experience.

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# REFERENCES

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# ANNEXURE 1

Annexure 1 contains supporting material that complements the main chapters of the BovineGuard project report but is too detailed to include in the core body. The information provided here helps readers, evaluators, and future developers to better understand the implementation details, configuration steps, and usage guidelines of the system. Material included in this annexure does not introduce new concepts; instead, it elaborates on existing modules, workflows, and testing described earlier.
Typical contents of Annexure 1 for BovineGuard are:

- **Detailed UI screenshots** for each user interface layer including the public-facing Landing Page, the Secure Authentication Portal for veterinarians/farmers, and the Real-time Monitoring Dashboard showing live telemetry and AI analysis.
- **Communication Protocol Specifications**, including the complete list of AT Commands used for the SIM800L GSM module (e.g., SMS dispatch, signal quality checks) and the JSON data payload structure used for transmitting physiological parameters (Temperature, Pulse, Motion) over the network.
- **Data Logging Structures** showing key fields for Cattle Nodes, Health Anomalies, Alert Logs, and User Profiles, illustrating how the system organizes historical data for trend analysis.
- **Firmware Configuration Details**, such as the pin_mapping.h file defining ESP32 S3 GPIO connections to sensors (DS18B20, MAX30100, MPU6050), and a list of required libraries illustrating how to set up the development environment.
- **Extended test case tables**, including edge cases for sensor calibration failures, battery low-voltage behaviors, and network timeout scenarios, beyond those summarized in the System Testing chapter.
- **User manual or hardware deployment guide** with step-by-step instructions for attaching the collar to the animal, interpreting LED status indicators on the node, charging the Li-ion battery, and registering a new Cattle Node ID in the system.

These additional resources make the project easier to replicate, evaluate, and extend, providing practical guidance for future students, researchers, and practitioners who wish to build on BovineGuard.